Malachy Kennedy

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SUMMARY

Game and software engineer with 2 shipped Steam titles and a third approved for release. Builds gameplay, combat and AI in Unreal Engine 5 (C++, GAS) and Unity 6 (C#), plus kernel, search and LLM tooling. Co-founder of Burnt Out Games, LLC and President of North America's largest collegiate game development club.

EDUCATION

University of California, Irvine Sep 2023 – Present
Donald Bren School of Information and Computer Sciences
B.S. Game Design & Interactive Media · B.S. Software Engineering · Major GPA 3.73 · Dean's Honor List

- Relevant Coursework: Games Engineering · C++ Programming · Java Programming · Data Structures & Algorithms · Linear Algebra · Probability & Statistics
- Also: Advanced Game Design · HCI · Software Engineering · Software Design · Information Retrieval · Operating Systems · Multiplayer Game Development · Interactive Storytelling · Games & Emotion

EXPERIENCE

Game Engineer & Co-Founder Apr 2024 – Present
***Burnt Out Games, LLC** · Unreal Engine 5, Unity 6, C++, C# · 10-developer studio*

- Co-founded a 10-developer studio; released 2 games to Steam in UE 5.3 and Unity 6, with a third UE 5.6 title approved.
- Technical Director across studio projects, owning engine versions, source control, and build pipelines.
- Lead programmer on the studio's Unreal titles, setting the C++ architecture other developers build against.
- Administer Perforce and engine upgrade paths (UE 5.3 to 5.8) across concurrent projects and teams.
- Drive Steam release engineering: packaging, graphics settings, platform requirements, and hotfixes.
- Coordinated legal formation, storefront setup, and publishing logistics for a studio shipping to real release deadlines.
- Built the studio's website with Claude Code: a static site on Cloudflare Pages with a Git-backed CMS.

Intern Game Developer Apr 2025 – Mar 2026
***Taralumen Games** · Unity 6, C#, Trello · CORA, 2.5D metroidvania* · Code Samples

- Delivered 40+ playable Unity builds for twice-weekly reviews, iterating mechanics on direct CEO and board feedback.
- Prototyped 30+ production-ready mechanics across 3 environments, now a co-development studio's reference build.
- Architected a 6,208-LOC C# codebase (67 scripts, 31 prefabs, 15 scenes) on SOLID, Strategy Pattern, and Dependency Injection.
- Engineered depth-based collision via layer bitmasking and raycasts to resolve a multi-plane 2.5D gamespace.
- Built a generic utility layer with C# Generics, LINQ, and Func delegates, replacing slow framework object lookups.
- Implemented a coroutine manager with sentinel abort logic for deterministic lifecycle and memory control.
- Designed 3 levels around the mechanics they introduce, so players learn each one by playing.
- Managed sprints, task tracking, and system validation in Trello to hold co-development studio deadlines.

Course Reader, Game Engineering Apr 2025 – Present
***UC Irvine, Donald Bren School of ICS** · Irvine, CA*

- Read five upper-division courses: Coding in Game Engines, Advanced Game Design (TTRPG), Multiplayer Game Project, Game Engineering, and Game Design Capstone.
- Grade C++, multithreading, networking, and engine-architecture work for ~100 students per term.
- Support students building in Unreal Engine, Unity, and Godot, evaluating networking, AI navigation, VFX, and rendering work.

LEADERSHIP

President (since 2026), prev. Vice President (2025), Programming Officer (2024) Jan 2024 – Present
***Video Game Development Club at UC Irvine** · vgdc-uci.com*

- Grew the club into North America's largest collegiate game dev organization: 600+ members, 44 officers.
- Increased funding 106%, engagement 104%, attendance 57%, Q1 attendance 71%, and acquisition 25%.
- Rewrote the constitution from 26 to 51 pages, defining output and securing the 106% funding increase.

- Onboarded a board 80% new after 30 of 39 officers graduated, rebuilding onboarding, applications, and interviews across 13 departments.
- Scaled a 3-officer graphics team into a 9-officer marketing department; authored a brand messaging guide and Figma style guide.
- Plan 6 large-scale events a year from 20 to 400+ attendees, plus 20 general meetings and 200+ workshops.
- Built 9 officer input sheets and Google Sheets API tooling for scheduling, attendance trends, and analytics.
- Mentored 5 teams of first-time developers, 3 in Unreal and 2 in Unity, through planning and production.
- Partner with UCI Esports on joint events, coordinating cross-club scheduling and communication.

GSET Program Mentor Jul 2025 – Aug 2025

***UCI GSET x Gifted and Talented Institute** · Unity · 33 students*

- Taught 33 high school students game development in Unity: scene building, player experience, and logic scripting.
- Mentored student teams through scoping, design, and debugging, taking each game from idea to a public showcase.

GAMES

Game Engineer & Tech Lead Aug 2025 – Present

***EXO: 3rd Person Speed Shooter** · Unreal Engine 5.6, C++, GAS · 11 developers · 3rd Place, ICS Project Expo · Steam*

- Sole programmer on an 11-developer title approved for release on Steam, integrating all art, audio, and design work.
- Engineered movement, feel, and camera: speed scaling, acceleration, camera shake, predictive look-ahead, and auto-targeting.
- Rebuilt Unreal's directional gravity for high-speed spherical traversal, using quaternion and vector math.
- Built a Risk of Rain 2-style spawn director allocating credits against weighted enemy cards by difficulty.
- Eliminated spawn hitches with an O(1) intrusive linked-list pool and a deferred queue capped to one alloc per tick.
- Engineered three enemy archetypes: predictive drones, arcing chargers, and an area-control boss that phase-shifts at 50% HP.
- Rewrote how GAS commits attributes, clamping through PreAttributeChange before values reach the attribute set.
- Split player logic across four tick groups to run physics before art and audio work each frame.
- Decoupled UI, audio, and VFX from gameplay through MVVM-style hooks, keeping UMG widgets reactive to state.
- Integrated Discord OAuth2 with PKCE, persisted tokens, and live rich presence; used World Partition and PCG for planets.
- Designed the resource economy: kills bank fuel spent on acceleration, so shooting well keeps the player fast.
- Ran three playtest rounds, lifting "dull to stimulating" 43% and attack satisfaction 36%; 200+ testers overall.
- Built the market research behind multiple publisher pitches, retargeting comparables and positioning to each catalog.
- Integrated Perforce, AWS EC2, UE 5.6, Rider, and Wwise into one pipeline; authored the onboarding workflow for 11 developers.

Combat Engineer & Designer Jun 2026 – Jul 2026

***The Fallen: Action RPG Combat Demo** · Unreal Engine 5.8, C++, GAS · UniJam 2026 (1 week), 8 developers · Portfolio*

- Engineered a full combat kit in one week: 4-hit chain, charged heavy, AoE special with charges, and a transformative ultimate.
- Built the animation system: cancel windows, input buffering, directional hitstop, i-frames, and AnimNotify hitboxes.
- Rewrote GAS's single-type damage pipeline in C++ for custom damage types, layered modifiers, and split callbacks.
- Modeled nine derived stats with formula-driven resolution, crit rolls, and armor-piercing math.
- Built the kit as one loop: attack speed feeds stagger, stagger opens armor pierce, and pierce returns energy for the ultimate.
- Built C++ targeting with anim warping, directional camera shake, and knockback; saved state with SaveGame serialization.

Technical Director & Narrative Designer Apr 2024 – May 2026

***Kawai'ian Isolation: 3D Narrative Horror** · Unreal Engine 5.3, Blueprints, Wwise · 30+ contributors · Steam*

- Shipped a UE5 title on Steam that reached 10,000+ claims in its first 100 hours (8 reviews, 100% positive).
- Led a technical department of 18+ contributors across audio, design, programming, and writing.
- Engineered a CSV-to-Unreal-to-Wwise pipeline for 56 pages of dialogue, branching playback on player choice and world state.
- Scripted mechanics, AI direction, save system, and sequencing for 22 scenes in Blueprints, toggling sections with Data Layers.

- Owned draw-call and LOD optimization: material instances, Nanite, Lumen, culling, streaming, and scalability settings.
- Wrote a 56-page game screenplay across 22 scenes with a second writer and two consulting narrative designers.
- Administered an AWS EC2 Perforce server and wrote the onboarding docs used by 30+ contributors.
- Managed cross-team pipelines with 10 custom Agile tools built in Miro and Google Sheets.

Gameplay & Network Engineer Jan 2025 – Aug 2025
Burning Out: Online Co-op Survival Horror *· Unity 6, C#, Mirror Networking · 7 developers ·* Steam
- Designed a networked inventory on Mirror with custom OOP containers and item templates, consistent across clients.
- Implemented movement and a torch-and-fire system with dynamic lighting, flame behavior, and torchwood burndown.
- Made finite, carried light the co-op cohesion mechanic, so players enforce grouping on each other.
- Ran in-person playtests with 100+ participants, converting clarity feedback into shipped changes.
- Published the studio's first Steam title: storefront setup, release logistics, and post-launch fixes.

Gameplay Engineer Oct 2026
Masks On: Online Stealth Arena Shooter *· Unreal Engine 5.8, Claude Code, Unreal MCP · Fall ZotJam 2026 (2 days), team of 2 ·* Itch
- Directed parallel Claude agents from written specs to a playable online FPS in 2 days on a team of 2.
- Reviewed every AI diff and tested each build by hand, catching agent bugs in recoil, sensitivity and HUD.
- Ran Epic's experimental UE 5.8 MCP plugin to validate a hand-built greybox level and test game-feel numbers.
- Integrated all gameplay into a teammate's Steam client-server netcode: up to 8 players, ~20 ms between clients.
- Prototyped the guns now used in a capstone shooter: recoil patterns, weapon sway, and aim-down-sights.

Engineer, Designer & Tech Artist Apr 2026
Night Walk: First-Person Survival Horror *· Unreal Engine 5.7 · ZotJam 2026 (72 hours), 5 developers ·* Portfolio
- Engineered first-person horror gameplay: movement, sprint, crouch, muffle, weapon equip/fire/aim, and camcorder zoom.
- Authored a post-process shader for camcorder night vision with power-on, flicker, filter, and zoom states.
- Designed UI in Figma and implemented it in-engine as UMG widgets; tuned step-synced shake and shotgun jolt.

Sole Programmer Jul 2025 – Aug 2025
Loonage: Custom Polar Physics Platformer *· Unity, C# · 96-hour jam, 6 developers*
- Built a custom 2D polar-coordinate physics system to replace Unity's built-in physics on a world of spinning disks.
- Engineered polar AABB collision with radial pushback and ground clamping for riding rotating platforms.

Sole Engineer Jan 2026
Little Rockstar: VR Narrative Experience *· Unity 6, C#, Meta XR SDK · Winter ZotJam 2026 (48 hours), 6 developers*
- Learned the Meta XR SDK from scratch and shipped physics interactions and event-driven story beats in 48 hours.

Gameplay Scripter Oct 2025
Trick or Treat: Roblox Multiplayer Game *· Roblox Studio, Luau · Fall ZotJam 2025 ·* Roblox
- Scripted a published multiplayer round-based game with randomized gravestone rewards and PvP candy stealing.

Sole Developer & Writer Sep 2025 – Dec 2025
LAPLACE: Interactive Speculative Fiction *· Twine, SugarCube, JavaScript ·* Itch
- Adapted an original short story into a 16-scene game with a diegetic typing interface and a CSV-driven text pipeline.

Game Designer & Engineer Apr 2025
Love Bomb: Dating Sim x Beat 'Em Up *· Unity, C#, Apps Script · ZotJam 2025 ·* Itch
- Tied combat strength to dialogue performance; built a Google Sheets CSV import pipeline for tagged, timed dialogue.

Gameplay & AI Engineer, Designer Aug 2024
Terrabloom: Ecosystem Builder *· Unity, C#, Cinemachine · GMTK Game Jam 2024 ·* Itch
- Engineered a plant-and-insect ecosystem on AI state machines, with ScriptableObject tuning and dual input modes.

Designer Nov 2024 – Jan 2025
PAPAWUDA: Game Dev Personality Quiz *· Twine, Google Forms, Google Sheets ·* Itch
- Mapped 8 trait dimensions to 8 club departments and 16 sub-roles with weighted scoring from a member survey.

SOFTWARE

Solo Developer Aug 2026 – Present
Resume System: Resume Pipeline & Job Watch *· Python, LibreOffice, ATS APIs, Claude Code ·* Portfolio

- Built an LLM-driven résumé pipeline (Claude Code, Python) that writes a tailored résumé for each posting from a verified master record.
- Gated every line against hallucination: each must trace to a recorded fact, past a never-claim list, locked titles, and a link whitelist.
- Stress-tested on 11 real postings, codifying each root-cause fix into a 400-line rules file every later run follows.
- Applied inverted-index concepts from my search engine, mapping keywords to entries so each run loads only relevant files.

Kernel Developer May 2026 – Jun 2026

Pintos: OS Thread Scheduler · *C, x86, GDB · Solo, optional course project* · Portfolio

- Engineered a preemptive thread scheduler in C with priority scheduling and a 4.4BSD multi-level feedback queue.
- Eliminated kernel timer busy-waiting with an ordered sleep list for O(1) wake-up at interrupt time.
- Mitigated priority inversion with nested donation to depth 8 across locks, semaphores, and condition variables.
- Built a fixed-point math library so the kernel runs scheduling math with no floating-point hardware.

Engineer Feb 2026 – Mar 2026

Search Engine & Web Crawler · *Information Retrieval · Python, BeautifulSoup, RegEx · Team of 4* · Portfolio

- Engineered a disk-based inverted index over 6,700+ pages using K-way external merge sort without exceeding RAM.
- Served sub-second queries from disk with byte-offset mapping; ranked with TF-IDF, cosine normalization, and Jaccard boosting.
- Designed a crawler with politeness delays and spider-trap defenses, deduplicating pages with MD5 hashing.

DESIGN & RESEARCH

Designer & UX Researcher Apr 2025

Limital: Abstracted Virtual OS Interface · *Figma · UCI Design-a-thon 2025 Finalist, 3 developers* · Devpost

- Cut self-reported sensory overload 37% across a 35-participant study, designing a focus-oriented virtual workspace.
- Ran multi-stage abstraction tests that cut high-overwhelm responses from 43% to 6% across the same cohort.

Designer, UX Researcher & Writer Apr 2026

Comments: Asynchronous XR Community Platform · *Figma, After Effects · UCI Design-a-thon 2026, 4 developers* · Portfolio

- Designed a spatial XR platform for AR glasses that anchors anonymous, asynchronous notes to real-world places.
- Surveyed 22 undergraduates; designed a constellation UI where reply threads form star patterns in 3D space.

SKILLS

Languages: C++ · C# · C · Python · Java · JavaScript · SQL · HLSL · Luau · UE5 Blueprints · Google Apps Script

Engines: Unreal Engine 5 (5.3 to 5.8) · Unity 6 · Unity WebGL · Roblox Studio · Meta XR SDK

Unreal: GAS · Gameplay Tags · Enhanced Input · Behavior Trees · AnimNotify · Motion Warping · Niagara · UMG · World Partition · Data Layers · PCG · SaveGame

Unity: ScriptableObjects · New Input System · Mirror Networking · NavMesh · Cinemachine · Coroutines · LayerMask bitmasking

Systems: OOD · SOLID · Strategy Pattern · Dependency Injection · Object Pooling · Event-driven architecture · DS&A · Quaternions & vector math

Low-Level Systems: Multithreading · Synchronization primitives · Priority scheduling & MLFQS · Interrupt handling · Fixed-point arithmetic · Memory management

Networking: Steamworks multiplayer · Unreal replication · Client/server authority · State synchronization · Discord API · OAuth2 + PKCE

Audio & Tech Art: Wwise · FMOD hooks · Post-process materials · Camera systems · Draw-call & LOD optimization · Premiere Pro · After Effects

Tools & Production: Perforce (AWS EC2 admin) · Git · Jira · Confluence · Trello · Miro · Figma · Steamworks publishing · itch.io · Claude Code · MCP · Technical documentation